LOAN ELIGIBILITY PREDICTION THROUGH PREDICTIVE MODELING

INTRODUCTION TO DATA MINING (DA-545)

FINAL-PROJECT

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Abstract

This project focuses on predicting loan approval outcomes for Sunrise Financial Services using machine learning models. Customer data, including demographics, income, education, employment status, and credit history, was utilized to build predictive models. Various machine learning algorithms were developed, tuned, and evaluated to assist in automating and optimizing the loan decision-making process. The project highlights the importance of data preprocessing, exploratory data analysis (EDA), feature engineering, model selection, and hyperparameter tuning to achieve reliable performance in predictive modeling.

Keywords: Loan Eligibility Prediction — Machine Learning — Random Forest — XGBoost — Hyperparameter Tuning — Financial Risk Assessment

Introduction

Sunrise Financial Services specializes in offering personal and home loans across metropolitan, suburban, and rural regions. Loan applications are initiated through an online portal, requiring rapid eligibility assessment to maintain service competitiveness. Traditional manual processes are inefficient and susceptible to human bias, motivating the adoption of machine learning techniques to automate and enhance the loan decision-making process.

Sunrise Financial Services specializes in offering personal and home loans across metropolitan, suburban, and rural areas. Customers apply through an online portal, where the company evaluates their eligibility based on submitted information.

This project focuses on applying data mining techniques to automate the loan eligibility process, improve decision-making, and optimize marketing strategies.

Data Background

The dataset consists of applicant-level information from Sunrise Financial Services, collected through online loan applications. It includes both demographic and financial indicators necessary for evaluating loan eligibility.

Data Description

The training dataset contains 16 attributes, while the testing dataset contains 15 attributes. Key features include:

- Categorical Variables: Gender, Married, Dependents, Education, Self_Employed, Property_Area
- Numerical Variables: ApplicantIncome, CoapplicantIncome, LoanAmount, Loan_Amount_Term, Credit_History, TotalIncome, IncomePerPerson, LoanToIncomeRatio, LoanTermYears
- Target Variable: Loan_Status (0 = Not Approved, 1 = Approved)

Data Preprocessing

The biggest task in any Data Mining problem is Data Preprocessing. Before we apply any algorithms, we first need to clean the data. The data might contain missing values, outliers, etc. There could be different kinds of attributes in the given data, like continuous variables, categorical variables, nominal, ordinal, ranges, ratios, etc. Such variables need to be taken care

of based on the model we want to apply. Data preprocessing not only deals with these thing but also covers feature selection, data reduction, etc. Given a set of attributes, not every attribute is important for data analysis or prediction. Few attributes might be of no use and few attributes will be very vital for our analysis. For this purpose we do feature selection.

Missing Data Imputation

- For categorical variables such as Gender, Married, Dependents, Self_Employed, and Credit_History, the missing values were replaced using the mode (most frequently occurring value) of each respective attribute.
- For numerical variables such as LoanAmount and Loan_Amount_Term, the missing values were replaced using the median of existing values. This approach was selected to minimize the effect of outliers and skewed distribution

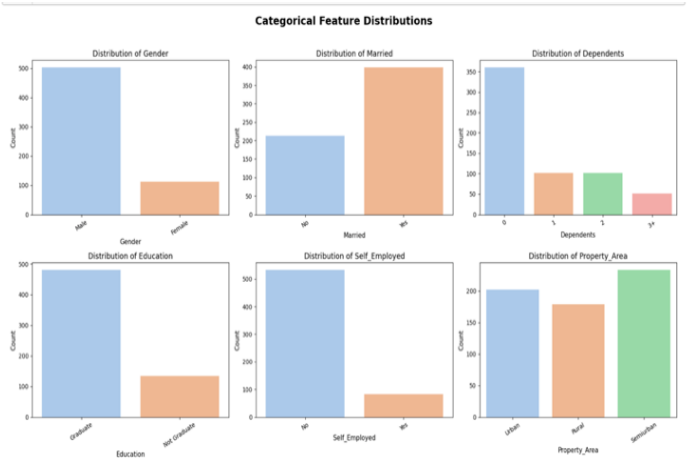
Validation After Imputation:

- Following imputation, checks were conducted to confirm that there were no missing values left in either the training or testing datasets.

This imputation method preserved the data's integrity and avoided introducing bias, ensuring that the dataset was complete and ready for reliable model training and testing.

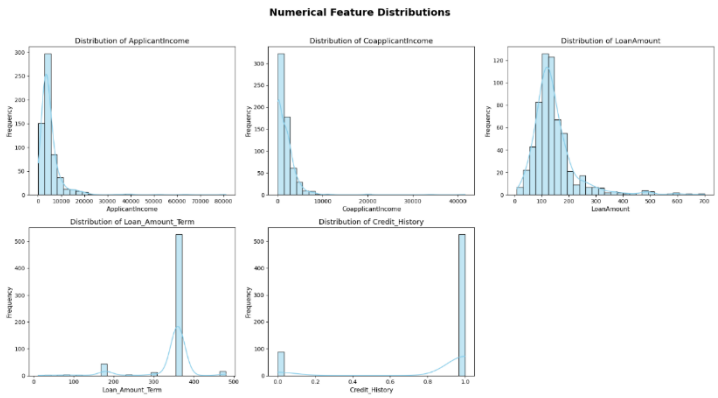
Exploratory Data Analysis (EDA)

To perform a comprehensive exploratory data analysis (EDA), we merged the training and testing datasets into a single combined dataset.
Merging the datasets allows us to analyze the complete distribution of features without being restricted to just the training set. This approach ensures that the EDA captures patterns, trends, and potential anomalies present across the entire data collection, providing deeper insights for feature engineering and modeling decisions.It is important to note that while merging is useful for visualization and understanding data characteristics, the test data's target variable (Loan_Status) remains unknown and is not used during model training or evaluation to avoid data leakage.



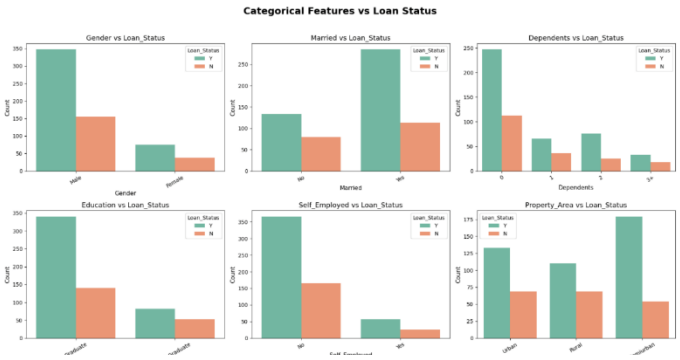
An analysis of the categorical features reveals key patterns in the dataset:
Gender: The majority of applicants are male, indicating a gender imbalance.
Marital Status: Most applicants are married, suggesting that marital status may play a role in loan applications.
Dependents: A significant portion of applicants have no dependents, while fewer have one, two, or three or more dependents.
Education: Most applicants are graduates, showing a strong representation of higher educational attainment.

Self-Employment: The majority are not self-employed, implying that most applicants have salaried jobs.
Property Area: Semiurban areas have the highest number of applicants, followed byurban and rural areas.
These distributions suggest that the dataset is skewed towards married, male, and educated individuals who are not self-employed and predominantly from semiurban areas. Understanding these trends is crucial for building unbiased and effective predictive models.

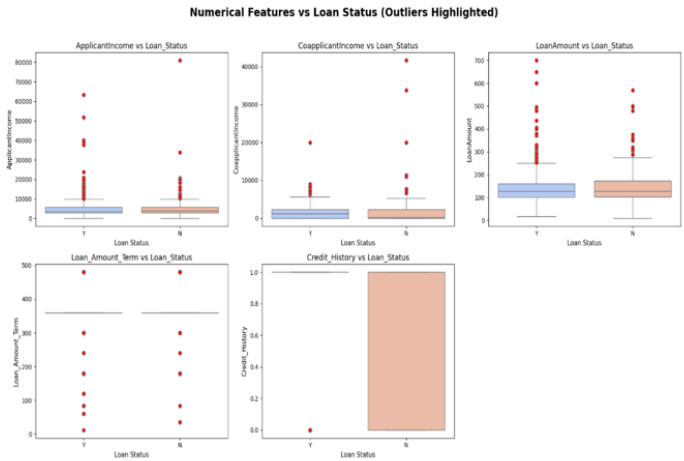


- Most applicants have low to moderate income; few have very high income.
- Coapplicant incomes are mostly low, with some outliers.
- Loan amounts are generally 100–200 units; few large loans exist.
- Loan terms mostly span 360 months (30 years).
- Most applicants have a good credit history (value = 1).

The graph shows the relationship between categorical features and loan approval status.Males and married applicants tend to have a higher number of approved loans. Applicants with no dependents have a better chance of loan approval. Graduates are more likely to get loans approved than non-graduates. Not self-employed applicants have a much higher loan approval rate than self-employed ones. Loans are approved more frequently for applicants from semiurban areas compared to urban or rural locations.



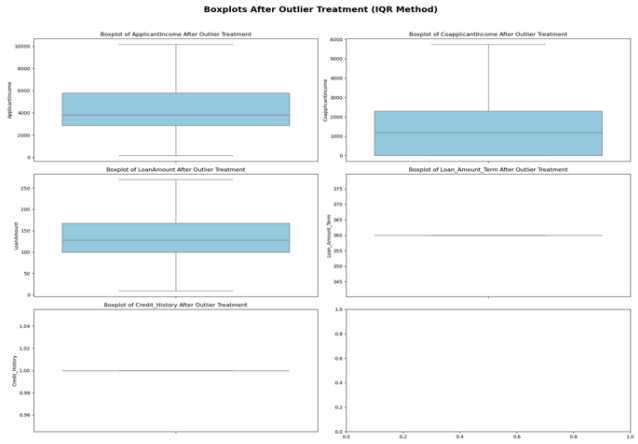
The graph shows the relationship between numerical features and loan approval status, with outliers highlighted. Applicants with lower ApplicantIncome and CoapplicantIncome generally have ahigher loan approval rate, while very high incomes (outliers) are less common.Approved loans tend to have lower LoanAmounts compared to rejected ones.Loan_Amount_Term is mostly similar across approval statuses, but most approved loans



have standard terms around360months.Credit_History strongly impacts loanstatus: applicants with a good credit history (value = 1) are much more likely to get loan approval.

Missing value and outlier treatment

To improve data quality and ensure more reliable model training, we applied outlier treatment to the key numerical features using the Interquartile Range (IQR) method.



The IQR method identifies extreme values that fall outside the typical range and adjusts or caps them to minimize their impact on the model.

Outlier treatment was performed for the following features:

- **ApplicantIncome:**
  Extremely high income values were capped to reduce the influence of very large incomes that could skew model training**.**
- **CoapplicantIncome:**
  Similar treatment was applied to coapplicant income, ensuring that a few extreme cases do not distort overall patterns.
- **LoanAmount:**
  Very large loan amounts were adjusted to fall within a reasonable range, making the dataset more balanced and improving prediction stability.
- **Loan_Amount_Term:**
  Loan terms that were unusually short or long were treated to align with typical term lengths seen in standard loan products.
- **Credit_History:**
  Outlier handling ensured that any anomalies in the credit history field, such as invalid or extreme values, were corrected to reflect valid credit behaviors.

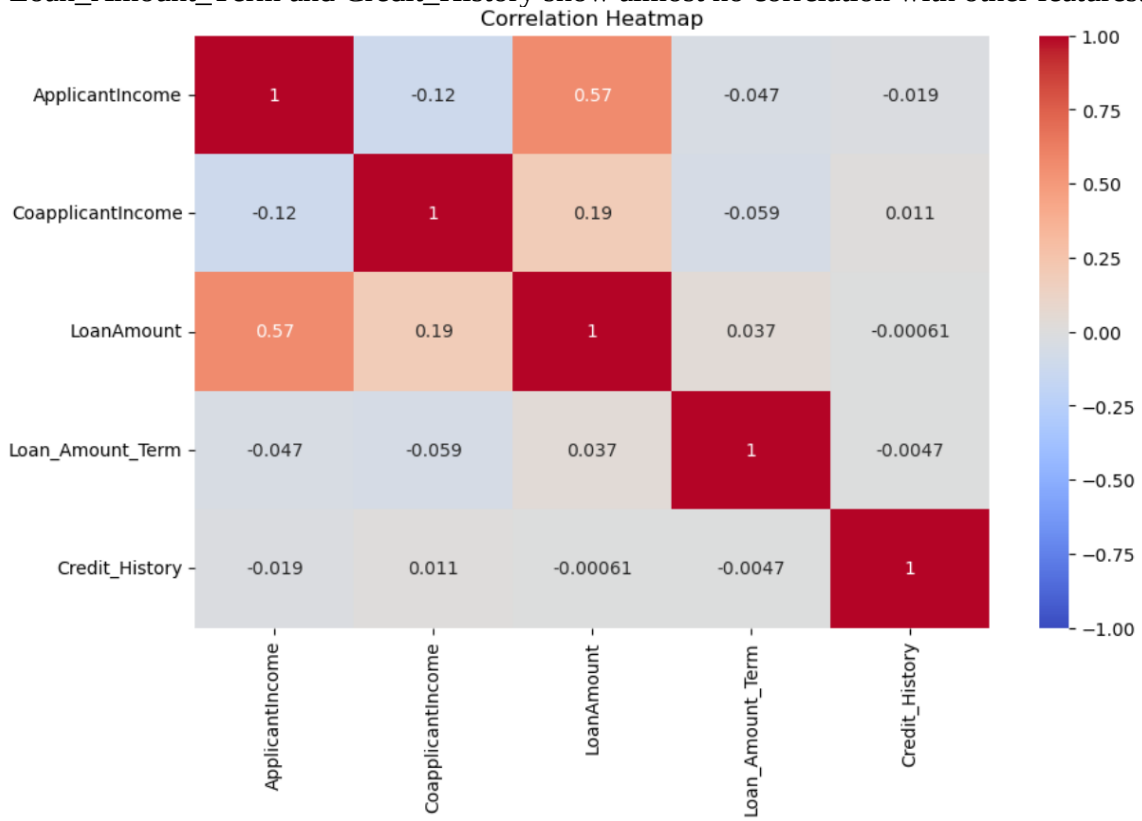
Correlation study: Heat map

This heatmap displays the pairwise correlation coefficients between important numerical variables.

The color and the number indicate how strongly two variables move together:

- Positive correlation (closer to +1) means that as one variable increases, the other also tends to increase.
- Negative correlation (closer to -1) means that as one variable increases, the other tends to decrease.
- ApplicantIncome and LoanAmount have a moderate positive correlation (0.57). Higher income applicants tend to request larger loan amounts.
- CoapplicantIncome has a weak positive correlation with LoanAmount (0.19).

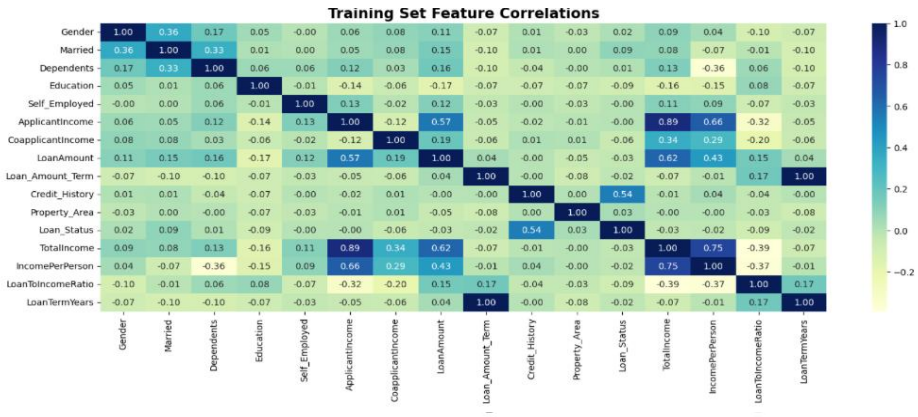
- Loan_Amount_Term and Credit_History show almost no correlation with other features.

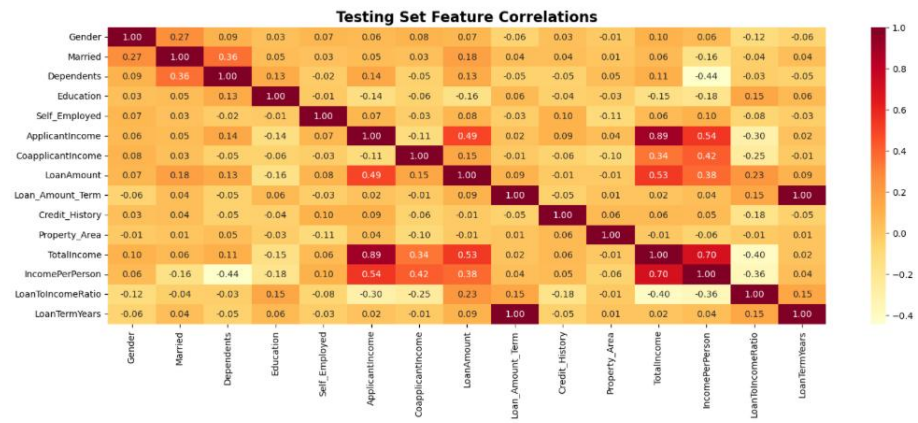


Feature Engineering

- In the feature engineering step, we created new features and improved the dataset structure to enhance model performance:
- **TotalIncome** was created by summing ApplicantIncome and CoapplicantIncome to better capture an applicant's overall financial capability.
- **Dependents** values were cleaned by converting '3+' into 3 and ensuring all entries were numeric, making the data suitable for machine learning models.
- **IncomePerPerson** was calculated by dividing TotalIncome by (Dependents + 1), providing insight into the financial burden on the applicant.
- **LoanToIncomeRatio** was introduced by dividing LoanAmount by TotalIncome, helping measure the proportion of the loan relative to the applicant’s income.
- **LoanTermYears** converted the Loan_Amount_Term from months to years, making loan durations easier to interpret.
- In addition to feature creation:
- The **Loan_ID** column was dropped, as it served only as an identifier and had no predictive value.
- The target variable **Loan_Status** was encoded from categorical ('Y', 'N') into numeric (1, 0) for model compatibility.
- Missing values in both training and testing datasets were handled properly, resulting in complete, clean datasets.

Training Set:
The heatmap shows that **Credit_History** is strongly positively correlated with **Loan_Status**, meaning applicants with good credit are more likely to be approved. **TotalIncome** and **LoanAmount** are moderately correlated, showing higher income applicants tend to request larger loans.





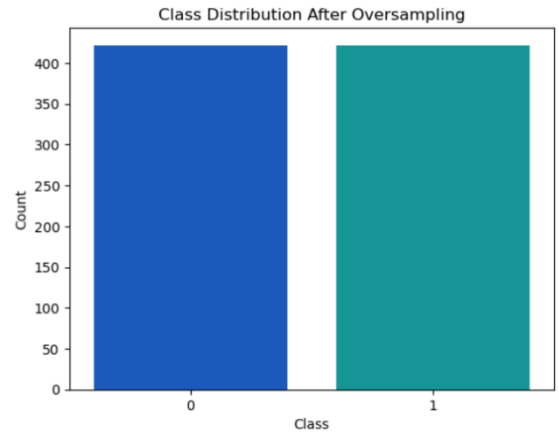
Explanation of Correlation Heatmaps

Testing Set:
The heatmap shows similar patterns as the training set. **TotalIncome** is positively correlated with **LoanAmount**, and **LoanTermYears** is perfectly correlated with **Loan_Amount_Term**, confirming the consistency of feature relationships across datasets.

MODEL BUILDING

First, the feature matrix X and target vector y were prepared. X was created by dropping the Loan_Status column from the training dataset, while y was set as the Loan_Status values. This separation was necessary to clearly define the input features and the target variable for the machine learning model.

To address the class imbalance problem, RandomOverSampler from the imblearn library was applied. It balanced the dataset by randomly duplicating samples from the minority class untiboth classes had equal representation. Afte r oversampling, a count plot showed that the distribution of both classes was now equal. This step was important to prevent model bias toward the majority class and to improve overall prediction accuracy.



In this project, we built and evaluated four machine learning models: **Logistic Regression**, **Random Forest**, **Decision Tree**, and **XGBoost**. Each model's performance was assessed using **accuracy**, **confusion matrices**, **classification reports** (precision, recall, f1-score), and **cross-validation scores** to ensure a thorough evaluation.

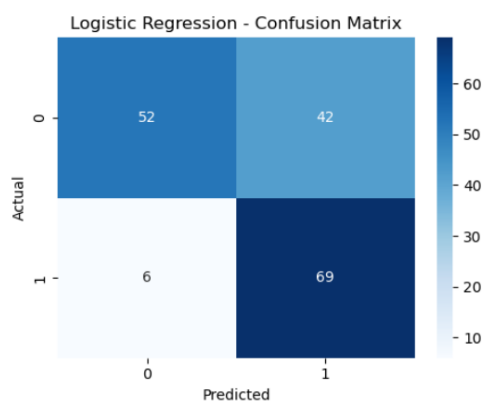
Logistic Regression

- **Accuracy:** 71.60%
- **Confusion Matrix:** The model pe rformed well at identifying loan a pprovals (class 1) but struggled w ith loan rejections (class 0), as sho wn by a lower recall (0.55) for cla ss 0.
- **Classification Report:** Precision for class 0 was high (0.90), but m any negatives were incorrectly cla ssified as positives.
- **Cross-Validation:** The mean cros s-validation score was **70.72%**, in dicating that Logistic Regression was moderately stable but had roo m for improvement. Variations in cross-validation scores also sugge st sensitivity to data splits.

```
===== Logistic Regression =====
Accuracy: 0.7159763313609467
Confusion Matrix:
[[52 42]
 [ 6 69]]
Classification Report:
              precision    recall  f1-score   support

           0       0.90      0.55      0.68        94
           1       0.62      0.92      0.74        75

    accuracy                           0.72       169
   macro avg       0.76      0.74      0.71       169
weighted avg       0.77      0.72      0.71       169
```

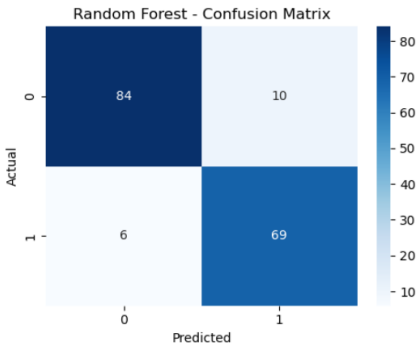



```
Cross-Validation Scores: [0.74117647 0.69411765 0.74117647 0.74117647 0.72619048 0.77380952
 0.69047619 0.6547619  0.66666667 0.64285714]
Mean Cross-Validation Score: 0.7072408963585435
```

```
===== Random Forest =====
Accuracy: 0.9053254437869822
Confusion Matrix:
[[84 10]
 [ 6 69]]
Classification Report:
              precision    recall  f1-score   support

           0       0.93      0.89      0.91        94
           1       0.87      0.92      0.90        75

    accuracy                           0.91       169
   macro avg       0.90      0.91      0.90       169
weighted avg       0.91      0.91      0.91       169
```




```
Cross-Validation Scores: [0.94117647 0.82352941 0.91764706 0.87058824 0.9047619  0.95238095
 0.95238095 0.89285714 0.94047619 0.92857143]
Mean Cross-Validation Score: 0.9124369747899159
```

Random Forest

Accuracy: 90.53%
Confusion Matrix: The model showed balanced predictions for both loan approvals and rejections, with minimal misclassifications.
Classification Report: High precision (0.93) and recall (0.89) for class 0, and strong performance for class 1 (recall 0.92), indicating robustness.
Cross-Validation: The mean cross-validation score was **91.24%**, the highest among all models, with low variation between folds. This suggests Random Forest is highly reliable and generalizes very well across different subsets of the data.

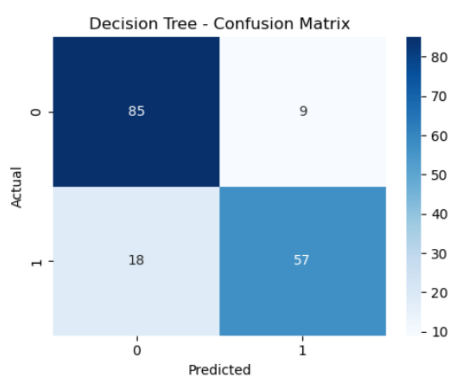
Decision Tree

- **Accuracy:** 84.02%
- **Confusion Matrix:** Decision Tree correctly classified most loan rejections (high recall for class 0 at 0.90) but made more mistakes on loan approvals (recall for class 1 was lower at 0.76).
- **Classification Report:** Precision and recall were well-balanced, though not as strong as Random Forest.
- **Cross-Validation:** The mean score was **87.56%**, showing fairly strong but slightly more variable performance compared to Random Forest.

```
===== Decision Tree =====
Accuracy: 0.8402366863905325
Confusion Matrix:
[[85  9]
 [18 57]]
Classification Report:
              precision    recall  f1-score   support

           0       0.83      0.90      0.86        94
           1       0.86      0.76      0.81        75

    accuracy                           0.84       169
   macro avg       0.84      0.83      0.84       169
weighted avg       0.84      0.84      0.84       169
```

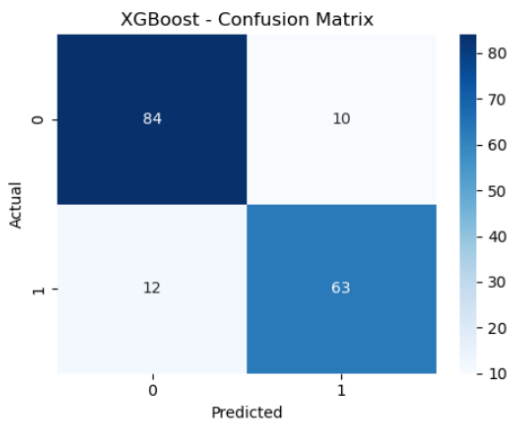



```
Cross-Validation Scores: [0.89411765 0.82352941 0.88235294 0.85882353 0.85714286 0.9047619
 0.9047619  0.9047619  0.8452381  0.88095238]
Mean Cross-Validation Score: 0.8756442577030812
```

```
===== XGBoost =====
Accuracy: 0.8698224852071006
Confusion Matrix:
[[84 10]
 [12 63]]
Classification Report:
              precision    recall  f1-score   support

           0       0.88      0.89      0.88        94
           1       0.86      0.84      0.85        75

    accuracy                           0.87       169
   macro avg       0.87      0.87      0.87       169
weighted avg       0.87      0.87      0.87       169
```




```
Cross-Validation Scores: [0.92941176 0.82352941 0.89411765 0.89411765 0.91666667 0.91666667
 0.96428571 0.88095238 0.91666667 0.94047619]
Mean Cross-Validation Score: 0.907689075630252
```

XGBoost

Accuracy: 86.98%
Confusion Matrix: XGBoost maintained a good balance between classes with relatively few misclassifications.
Classification Report: Both classes had high precision and recall (around 0.85–0.88), indicating balanced predictions.
Cross-Validation: The mean cross-validation score was **90.77%**, almost as high as Random Forest, suggesting strong predictive power and excellent generalization.

Why Cross-Validation Was Used

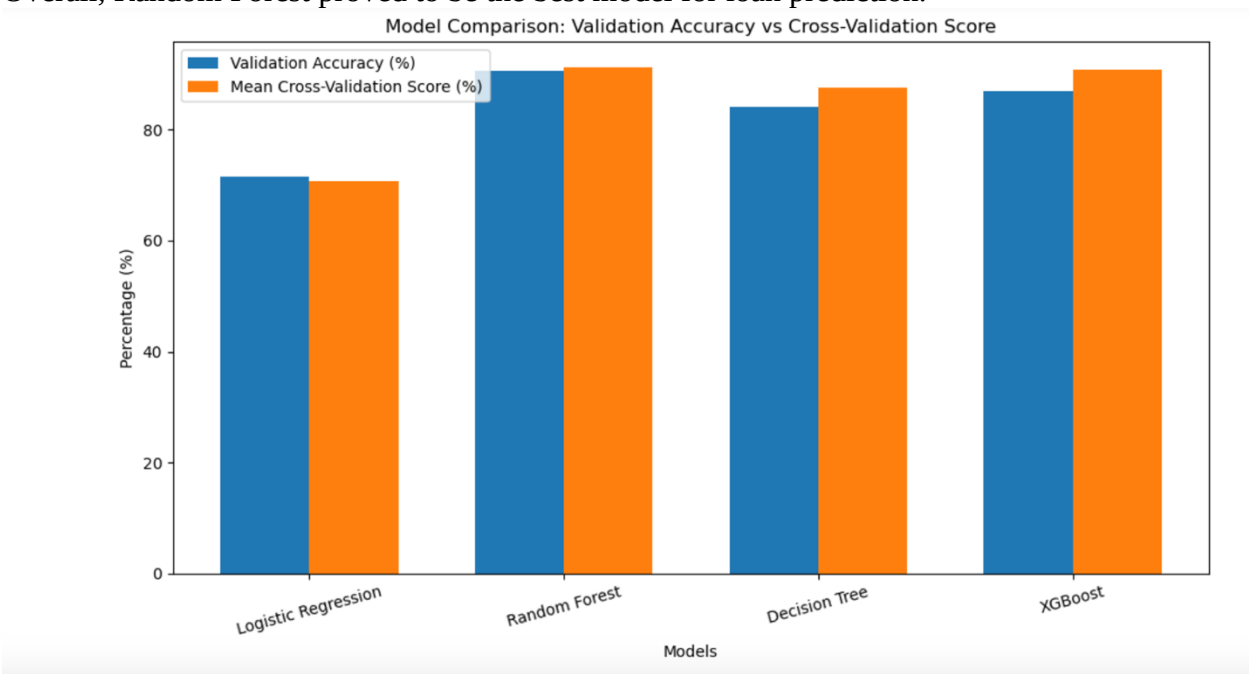
Cross-validation was a critical step in the evaluation process for several reasons:

- **Prevents Overfitting:**
  It tests the model on multiple unseen subsets of the data, ensuring the model does not just memorize the training set but generalizes well to new data.
- **Provides a More Reliable Estimate:**
  A single train-test split can be misleading if the split happens to favor the model. Cross-validation averages performance across different splits, giving a more dependable measure of model performance.
- **Assesses Stability and Robustness:**
  Models that perform consistently across multiple folds are considered more stable and reliable. Cross-validation scores help identify if a model is sensitive to certain data partitions.
- **Guides Model Selection:**
  By comparing mean cross-validation scores, we can confidently choose the best-performing model not just based on one lucky or unlucky split but based on its performance across many different subsets.

MODEL EVALVATION

Model	**Precision (0/1)**	**Recall (0/1)**	**F1-Score (0/1)**	**Overall Accuracy**	**Mean Cross-Validation Score**
Logistic Regression	0.90 / 0.62	0.55 / 0.92	0.68 / 0.74	71.60%	70.72%
Random Forest	0.93 / 0.87	0.89 / 0.92	0.91 / 0.90	90.53%	91.24%
Decision Tree	0.83 / 0.86	0.90 / 0.76	0.86 / 0.81	84.02%	87.56%
XGBoost	0.88 / 0.86	0.89 / 0.84	0.88 / 0.85	86.98%	90.77%

The bar chart compares validation accuracy and mean cross-validation scores for four models.
Random Forest achieved the highest and most consistent performance, with both scores above 90%.
XGBoost also performed strongly, closely following Random Forest.
Decision Tree showed good results but was slightly less stable, while **Logistic Regression** had the lowest accuracy and stability.
Overall, Random Forest proved to be the best model for loan prediction.

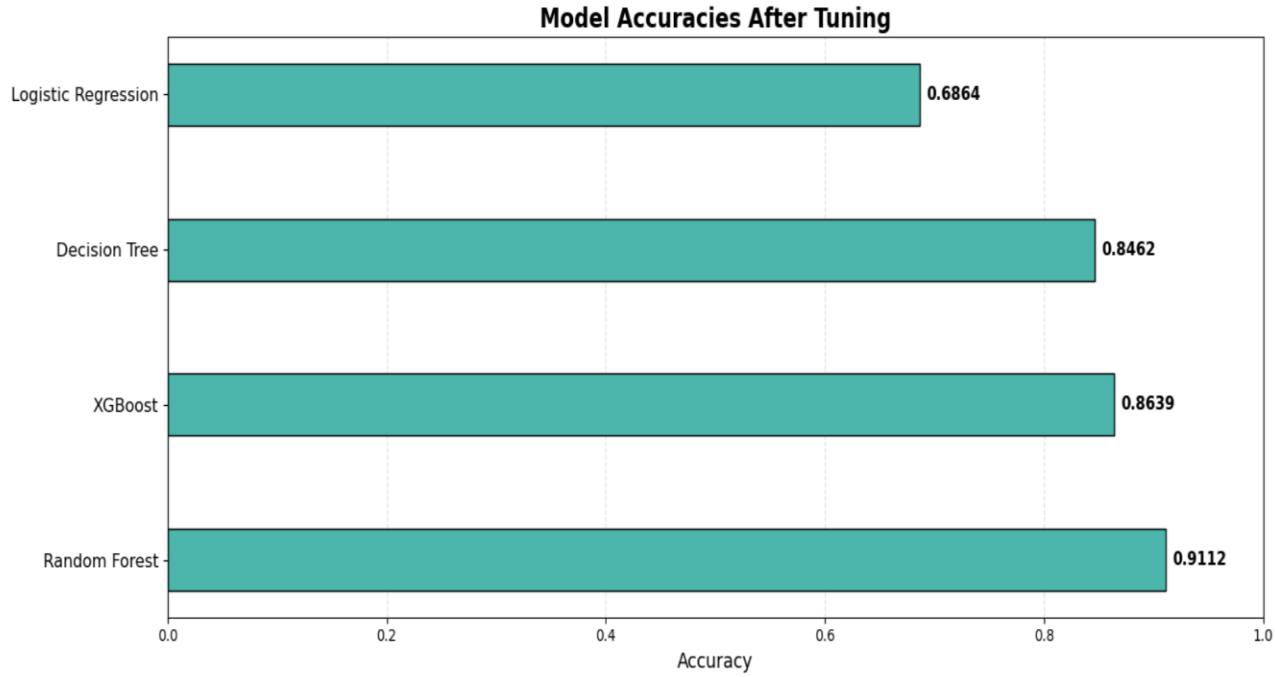


Hyperparameter Tuning

After initially building the models, we performed hyperparameter tuning for each of the four algorithms — Random Forest, XGBoost, Decision Tree, and Logistic Regression — to optimize their performance.
Hyperparameter tuning is crucial because machine learning models have adjustable settings (hyperparameters) that can significantly impact their accuracy and generalization ability. Finding the best combination of these parameters helps the models learn patterns more effectively and avoid overfitting or underfitting.

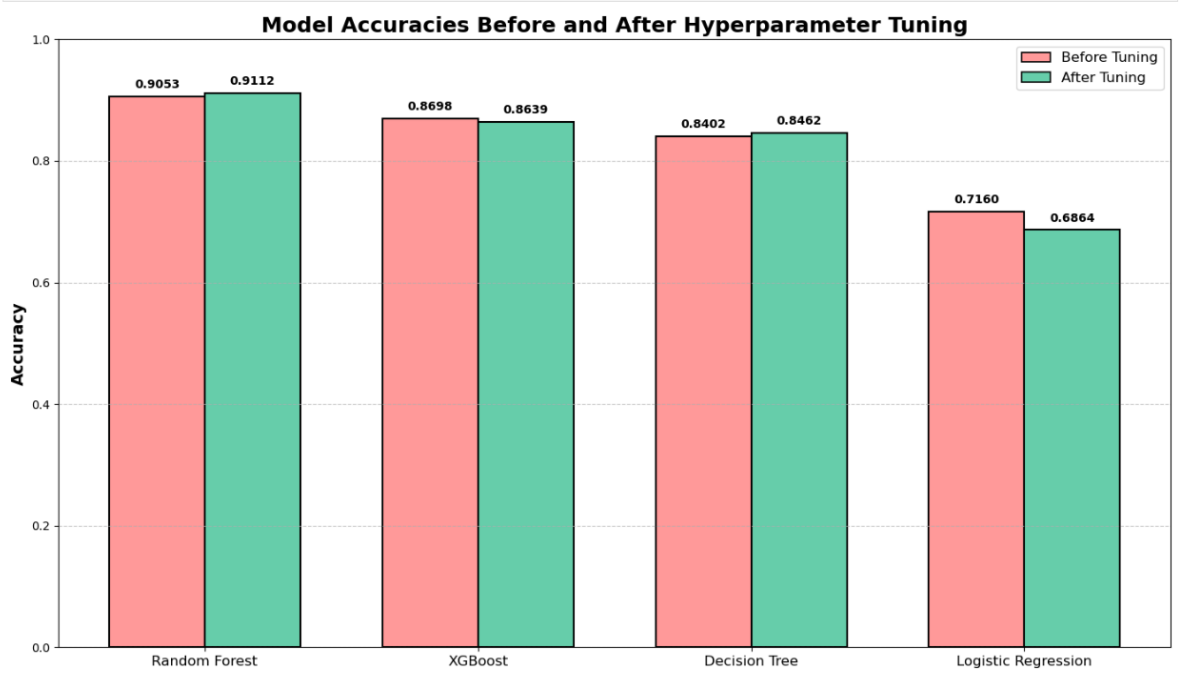
Model	**Best Parameters**	**Accuracy After Tuning**
Random Forest	bootstrap=False, max_depth=20, n_estimators=100	91.12%
XGBoost	colsample_bytree=0.8, learning_rate=0.2, max_depth=6, n_estimators=100	86.39%
Decision Tree	criterion='entropy', max_depth=None	84.62%
Logistic Regression	C=0.1, penalty='l1', solver='liblinear'	68.64%

We plotted a horizontal bar graph to clearly compare the **final tuned model accuracies**.
This visualization makes it easy to see which model performed best after hyperparameter optimization.



Model Accuracies Before and After Tuning

The chart compares model accuracies before and after hyperparameter tuning.
Random Forest and Decision Tree showed slight improvements after tuning, with Random Forest reaching the highest accuracy at 91.12%.
XGBoost and Logistic Regression had small drops in accuracy, indicating their default settings were already close to optimal.
Overall, tuning helped refine the models, with Random Forest remaining the best performer.



Results & Recommendations

In this project, we developed and optimized machine learning models to predict loan approval outcomes for Sunrise Financial Services. Four models were evaluated: Random Forest, XGBoost, Decision Tree, and Logistic Regression. Each model was fine-tuned using GridSearchCV to maximize predictive accuracy.

Random Forest achieved the highest accuracy, improving from 90.53% before tuning to 91.12% after tuning, demonstrating its strong ability to generalize and handle complex data.XGBoost maintained strong performance with an accuracy of 86.39% after tuning. Decision Tree slightly improved from 84.02% to 84.62%, indicating marginal gains with tuning. Logistic Regression decreased performance from 71.60% to 68.64%, reflecting its limitations in capturing complex patterns compared to ensemble models. Comparisons between models before and after hyperparameter tuning emphasized that ensemble methods (Random Forest and XGBoost) outperform simpler models, especially on structured financial datasets.

Usefulness for Decision-Making
The developed predictive models, especially the Random Forest, will help Sunrise Financial Services accelerate the loan approval process, improve customer experience, reduce financial risk, and ensure fair, consistent loan decisions. Machine learning enables faster, smarter, and more strategic operations by prioritizing high-quality applicants and refining marketing efforts.

Final Recommendation:
Random Forest is recommended for deployment due to its high accuracy, stability, and interpretability, making it ideal for both daily operations and strategic planning.

Future Work

- **Improve Feature Engineering:** Create new features like debt-to-income ratio for better predictions.
- **Use Advanced Oversampling:** Apply techniques like SMOTE instead of basic random oversampling.
- **Optimize Tuning Methods:** Use Randomized Search or Bayesian Optimization for faster, better hyperparameter tuning.
- Ensemble Learning: Combine models (e.g., Random Forest + XGBoost) to boost accuracy.
- **Model Explainability:** Use tools like SHAP or LIME to interpret model decisions.
- Deployment: Deploy the model into real applications and monitor performance over time.

Acknowledgement

We would like to express our sincere gratitude to Professor Minhee Jun for their valuable guidance, continuous support, and encouragement throughout this project.
Their insights and feedback were instrumental in helping us successfully complete the Loan Eligibility Prediction project.
We are truly thankful for the opportunity to apply the concepts learned in class to a real-world scenario.

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